BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLYPROGRESSREPORT (01/06/2026 – 07/06/2026)

WEEK 3

PROJECT NAME	RADIO PROPAGATION ANALYSIS AND PATH PREDICTION

PROJECT DESCRIPTION :	**PROJECT FOCUSES ON SIMULATING ANTENNA BEAM PROPAGATION**
GROUP MEMBER :	**GARIMA CHOUDHARY, MANASH PRASAR, ARYAN KUMAR**
GROUP ID :	**2026G265**
GROUP GUIDE :	**Dr. ABDUL JHUMMARWALA**
GROUP COORDINATOR :	**GARIMA CHOUDHARY**
COLLEGE GUIDE NAME :	**ABHILASHA SAKSENA**
COLLEGE GUIDE No. :	*81231 06272*
COLLEGE GUIDE EMAIL. :	**abhilasha.saksena@gsv.ac.in**
COLLEGE NAME :	**GATI SHAKTI VISHWAVIDYALAYA**

01/06/2026	• **Studied antenna radiation pattern fundamentals and gain representation concepts (Garima Choudhary)** • **Started designing workflow for 3D antenna radiation pattern visualization using CesiumJS and analyzed theta-phi-gain dataset structure (Manash Prasar)** • **Explored basics of antenna propagation characteristics and visualization requirements (Aryan Kumar)**
02/06/2026	• **Learnt coordinate transformation concepts used in antenna pattern visualization (Garima Choudhary)** • **Implemented spherical-to-Cartesian coordinate conversion for antenna radiation pattern points (Manash Prasar)** • **Assisted in understanding radiation pattern data organization and processing methods (Aryan Kumar)**
03/06/2026	• **Studied structured representation of radiation pattern datasets and gain distribution (Garima Choudhary)** • **Developed theta-phi grid generation logic for organizing antenna pattern samples in CesiumJS (Manash Prasar)** • **Explored methods for validating generated antenna pattern coordinates (Aryan Kumar)**
04/06/2026	• **Learnt mesh construction concepts and visualization techniques for 3D antenna patterns (Garima Choudhary)** • **Implemented triangular mesh generation using neighboring grid points to create continuous radiation surfaces (Manash Prasar)** • **Assisted in testing generated mesh connectivity and surface formation (Aryan Kumar)**
05/06/2026	• **Assisted in analyzing visualization outputs and identifying rendering anomalies (Garima Choudhary)** • **Performed debugging and optimization of gain scaling to reduce distortion in generated antenna lobes (Manash Prasar)** • **Participated in testing different rendering parameters and validating generated patterns (Aryan Kumar)**
06/06/2026	• **Reviewed generated radiation pattern outputs and supported final validation activities (Garima Choudhary)** • **Refined main lobe and side lobe visualization, improved overall 3D radiation pattern representation (Manash Prasar)** • **Assisted in result verification and documentation of visualization observations (Aryan Kumar)**

07/06/2026	Holiday (Sunday)

Next Week Plan	Next week, the focus will be on improving the accuracy and visual quality of the 3D antenna radiation pattern by refining the main and side lobes, optimizing mesh generation, and implementing gain-based color mapping. Further validation and testing will be carried out to ensure realistic radiation pattern representation, followed by documentation and preparation of the final project results.

REFERENCE:

https://octave.org/doc/interpreter/

https://www.antenna-theory.com/

https://www.qsl.net/4nec2/

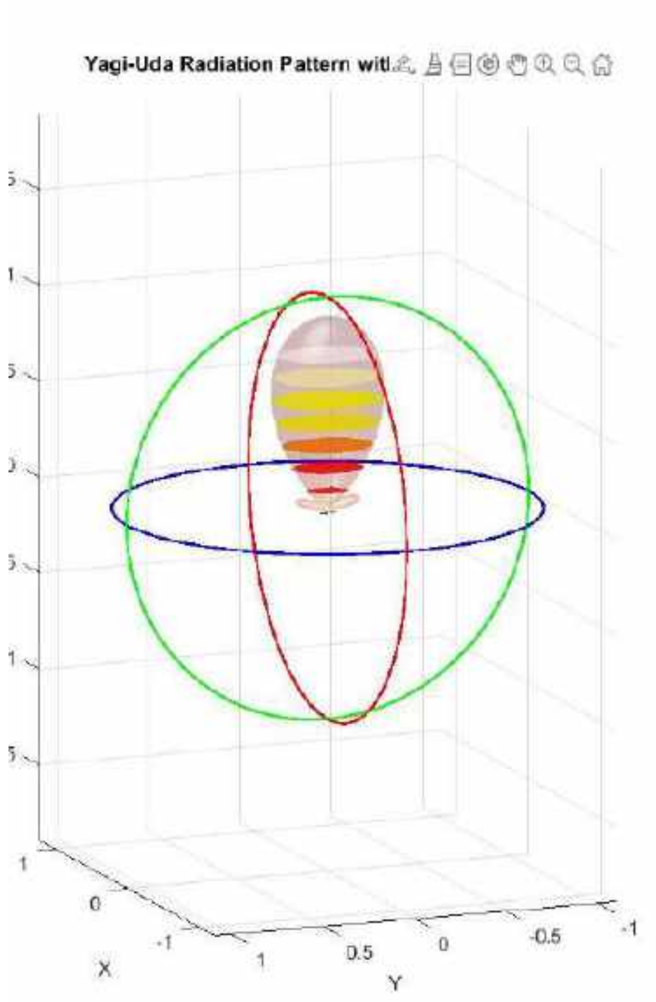
https://www.everythingrf.com/rf-calculators/link-budget-calculator

https://sandcastle.cesium.com/

https://www.qsl.net/kd2bd/splat.html

https://www.youtube.com/watch?v=ZaXm6wau-jc

SCREEN SHOTS:


Yagi-Uda Radiation Pattern witl
1
0
-1
X
1
0.5
0
-0.5
-1
Y